



Adaptive Collocation via Stochastic Solvers in PINNs

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OVERVIEW

Solving the Fokker-Planck Equation (FPE) in high dimensions is limited by the **curse of dimensionality**.

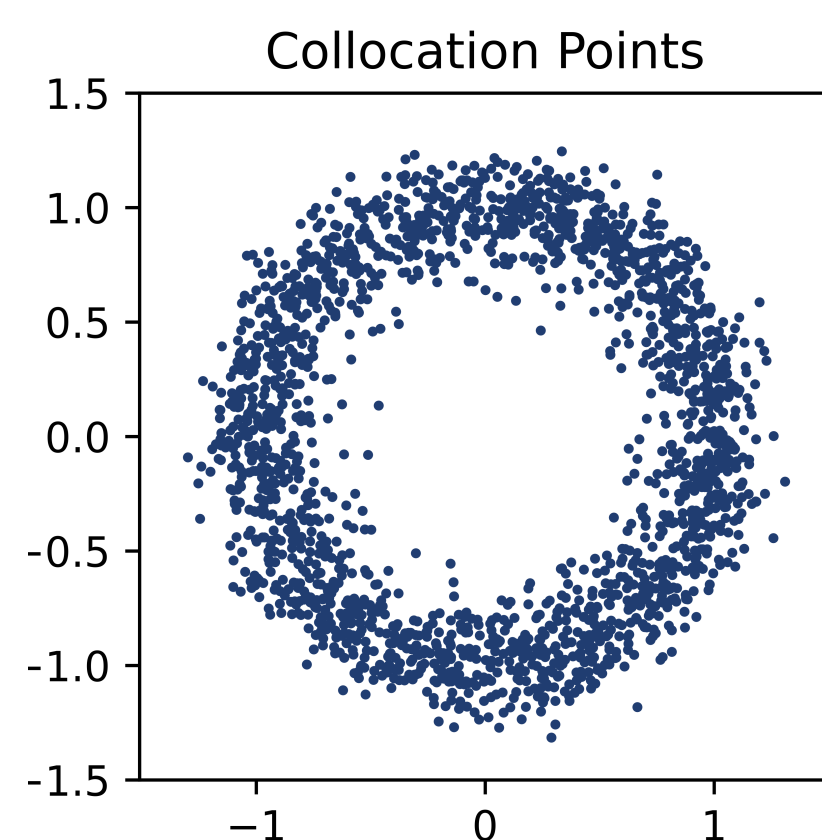
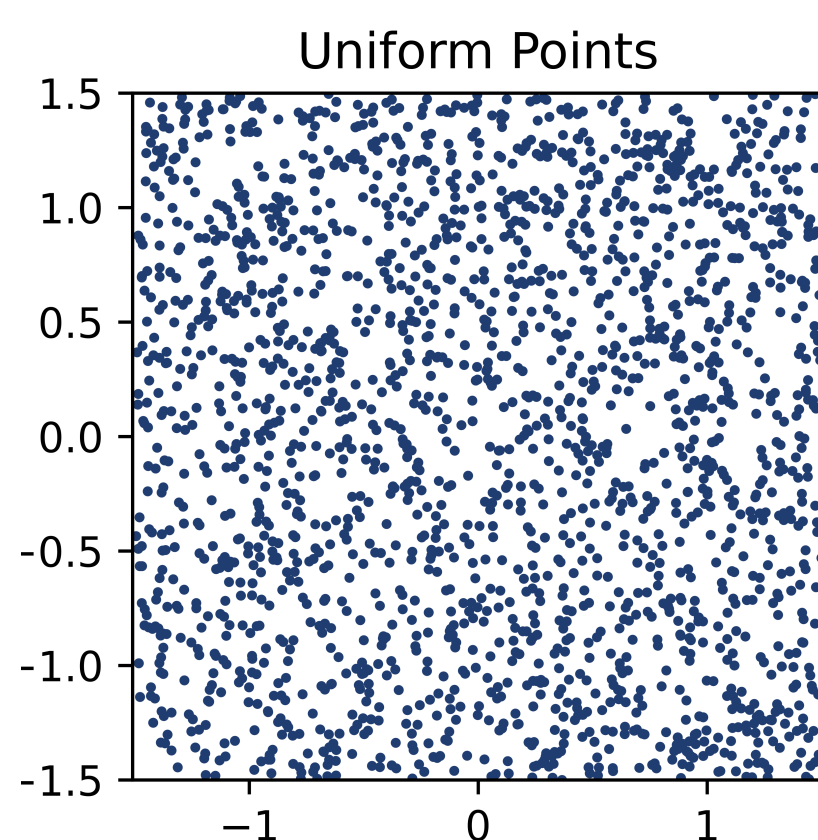
$$\text{FPE: } \partial_t u = \nabla_x \cdot \left(-fu + \frac{1}{2} \nabla_x \cdot (\Sigma u) \right)$$

$$\text{SDE: } dX_t = f(X_t, t)dt + \sigma(X_t, t)dW_t$$

The Core Idea

We propose a **mass-aware** PINN training strategy:

- **SDE-Driven:** Sampling via Euler-Maruyama to target high-probability regions (where the solution “lives”).
- **Importance Sampling:** Weighted physics loss for efficient optimization.
- **Hybridization:** Stochastic + Uniform points to capture tails and boundary layers.



RESULTS

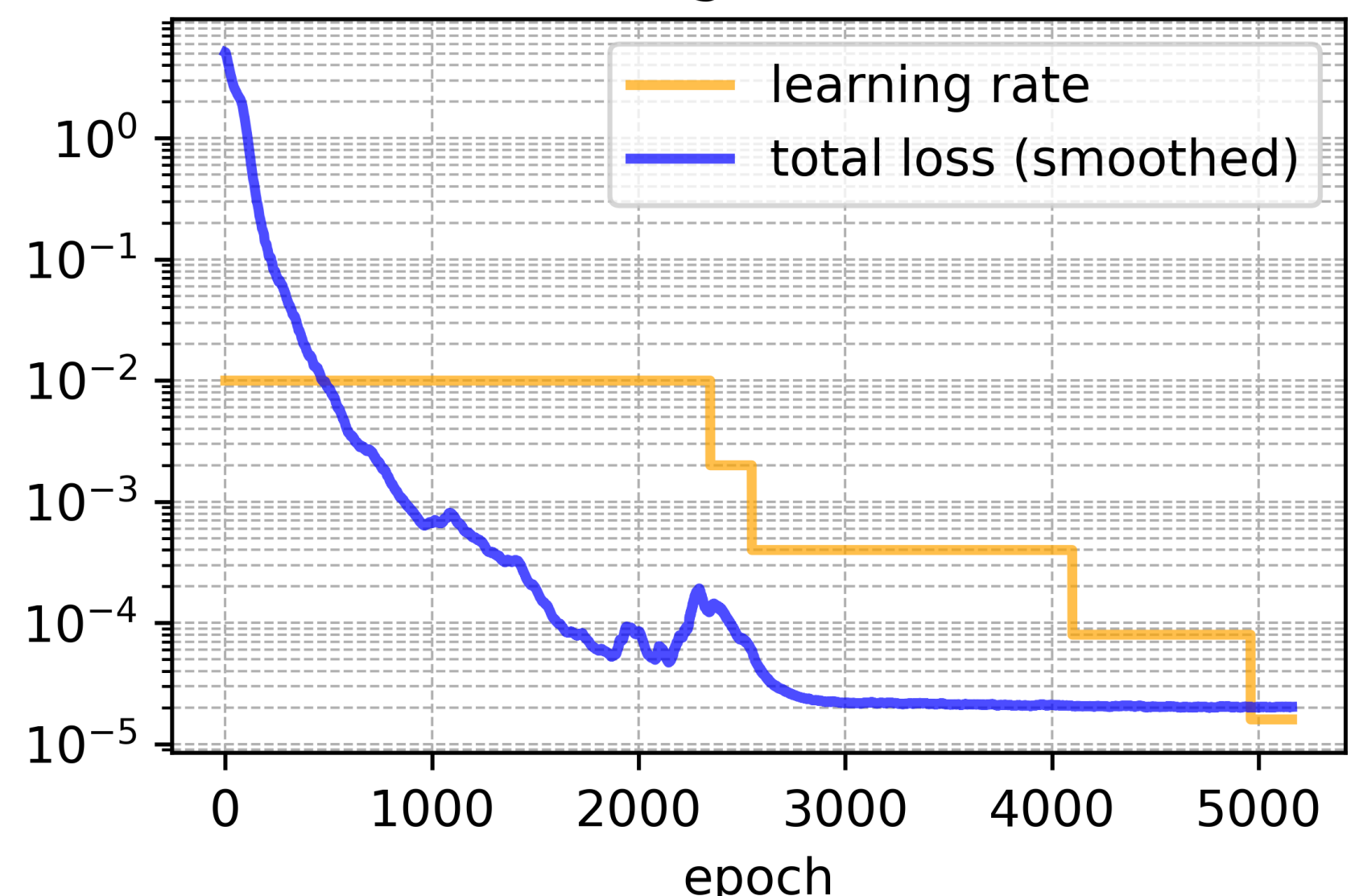
Numerical Setup & Key Findings

- **MLP Architecture:** (2,100,100,100,100,1)
- **Activation:** SiLU
- **Optimizer:** Adam

Our **Adaptive Collocation** vs. standard PINN (residual only):

- **Accuracy:** Slight reduction ($\sim 10\%$) in the KL divergence.
- **Stability:** Stable training runs and reduced overfitting.
- **Efficiency:** Training epochs reduced by 3%.

Training Overview



METHODOLOGY

Constrained Architecture

$u_\theta(x)$ is **positive** and L_1 -integrable by design:

$$u_\theta(x) = \text{SoftPlus}(N_\theta(x) + c(\theta)) \varphi(x)$$

N_θ : Deep Neural Network.

φ : Gaussian-like decay function.

$c(\theta)$: Value constraint at a reference point x_0 .

Hybrid Sampling

Dynamic resampling of the training set $\mathcal{X}$:

- **Stochastic:** 10% SDE points for high-density regions.
- **Uniform:** 90% coverage for tails and low-density zones.

Training data are resampled every epoch to balance focusing on high-probability regions with preserving global coverage, so the PINN does not lose resolution in low-density areas.

CONCLUSIONS

We introduced a mass-aware PINN framework that overcomes the limitations of uniform sampling in kinetic problems.

Future Challenges

- **Gradient Regularization:** Using histograms to smooth the loss gradient.
- **Time Dependency:** Extending the architecture to non-stationary problems.
- **Informed φ :** Engineering the decay function via underlying SDEs.
- **Adaptive Tuning:** Dynamic ratio adjustment of training points.

References

[1] J. Zhai, M. Dobson, and Y. Li, “A deep learning method for solving Fokker-Planck equations,” in *Proceedings of the 2nd Mathematical and Scientific Machine Learning Conference*, 2022, pp. 568–597.